as *WebAuthn* (https://webauthn.guide) provide many of the same security benefits and are much easier to use.

11.4.1 How TLS certificate authentication works

The full details of how TLS certificate authentication works would take many chapters on its own, but a sketch of how the process works in the most common case will help you to understand the security properties provided. TLS communication is split into two phases:

1. An initial *handshake*, in which the client and the server negotiate which cryptographic algorithms and protocol extensions to use, optionally authenticate each other, and agree on shared session keys.
2. An *application data* transmission phase in which the client and server use the shared session keys negotiated during the handshake to exchange data using symmetric authenticated encryption.[2]

During the handshake, the server presents its own certificate in a TLS *Certificate* message. Usually this is not a single certificate, but a *certificate chain*, as described in chapter 10: the server's certificate is signed by a certificate authority (CA), and the CA's certificate is included too. The CA may be an intermediate CA, in which case another CA also signs its certificate, and so on until at the end of the chain is a root CA that is directly trusted by the client. The